


Matrix factorization algos:

- SVD
(singular value decomposition)
- PCA
- QR decomposition
- LU decomposition
- ALS
- iALS

"truncated"

$$R \approx U \Sigma V^T$$

$k \ll l = \text{rank}(R)$

$$U^T U = V^T V = I_k$$
$$r_{ij} = U_i^T \Sigma V_j$$
$$U^T U = I$$

$\Rightarrow U$ - orthogonal matrix

$$U^{-1} = U^T$$
$$U^{-1} = U^*$$

unitary

ALS (Alternating Least Squares)

$$\min_{X, Y} \sum_{(i,j) \in R} (r_{ij} - x_i^T y_j)^2 + \lambda \sum_i \|x_i\|^2 C_i + \lambda \sum_j \|y_j\|^2 C_j$$

X - user matrix, Y - item matrix

R

1	x	x	x	x	x	x	x
x	x	x	x	x	x	x	0
x	x	1	x	x	x	x	x
x	x	x	x	0	x	x	x
x	x	x	x	x	x	x	x

undefined

$$R \approx XY$$

$$R \in \mathbb{R}^{n \times k}, X \in \mathbb{R}^{n \times l}, Y \in \mathbb{R}^{l \times k}$$

l - embedding dimension $\frac{\partial x_i}{\partial x_i} = 0$

$$\begin{aligned} \argmin_{x_i} P &= \argmin_{x_i} \left[\sum_{(i,j) \in R} r_{ij}^2 - 2 \sum_{(i,j) \in R} r_{ij} x_i^T y_j + \sum_{(i,j) \in R} (x_i^T y_j)^2 + \right. \\ &\quad \left. + \lambda \sum_i (x_i, x_i) C_i + \lambda \sum_j (y_j, y_j) C_j \right] = \\ &= \argmin_{x_i} \left[-2 x_i^T \sum_{(i,j) \in R} r_{ij} y_j + \sum_{(i,j) \in R} x_i^T y_j y_j^T x_i + \lambda C_i x_i^T x_i \right] = \\ &= \argmin_{x_i} \left[-2 x_i^T \underbrace{\left(\sum_{(i,j) \in R} r_{ij} y_j \right)}_{B_i} + x_i^T \underbrace{\left(\sum_{(i,j) \in R} y_j y_j^T + \lambda C_i I \right)}_{A_i} x_i \right] \\ &\Rightarrow \argmin_{x_i} (-2 x_i^T B_i + x_i^T A_i x_i) \end{aligned}$$

$$\frac{\partial}{\partial x_i} (-2 x_i^T B_i + x_i^T A_i x_i) = 0$$

$$\frac{\partial}{\partial x_i} (-2 x_i^T B_i) = -2 B_i, \quad \frac{\partial}{\partial x_i} (x_i^T A_i x_i) = (A_i + A_i^T) x_i = 2 A_i x_i$$

A -symmetric

$$-2 B_i + 2 A_i x_i = 0$$

$$\Rightarrow x_i = A_i^{-1} B_i$$

iALS (implicit ALS)

Implicit ALS $\neq$ implicit feedback

$$A \in \mathbb{R}^{n \times k} : \\ \text{sparsity} \approx 0.99\%$$

$$\frac{\# \text{ training cells eALS}}{\# \text{ training cells iALS}} = 0.01$$

$$p_{ui} = \begin{cases} 1, & r_{ui} > 0 \\ 0, & r_{ui} \leq 0 \text{ or undefined} \end{cases} \quad \text{— preference}$$

$$c_{ui} = 1 + \alpha / |r_{ui}|, \quad \alpha \sim 10^5 \quad \text{— confidence}$$

$$\underset{x_i}{\operatorname{argmin}} \sum_{i,j} c_{ij} (p_{ij} - x_i^T y_j)^2 + \lambda \sum_i \|x_i\|^2 \hat{C}_i + \lambda \sum_j \|y_j\|^2 \hat{C}_j \quad \ominus$$

$\ominus$ EXERCISE

$$c_{ij} (p_{ij} - x_i^T y_j)^2 \quad \text{— iALS}$$

$$(r_{ij} - x_i^T y_j)^2 \quad \text{— eALS}$$